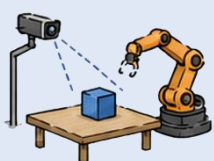


Why: Partial Observability, Competence Plateau, and Generalization Gap

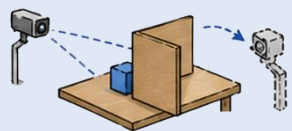
Imitation Policy



Embodied Explorer

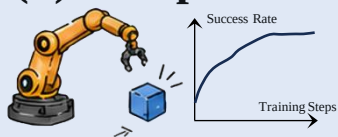


(a) Partial Observability



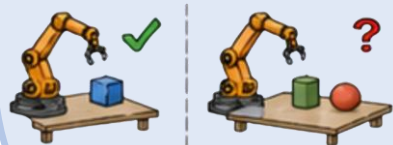
Reduce unseen spatial state, hidden physical attributes, and task ambiguity.

(b) Competence Plateau



Practice unmastered actions when imitation or static pretraining plateaus.

(c) Generalization Gap



Policies overfit to familiar tasks, narrow action modes, and single successful trajectories.

Where: From Perception to Embodied Action

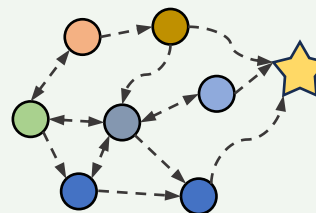
(a) Perception

- viewpoints / sensor poses
- maps / scene memory
- frontiers / occlusions
- object states / attributes
- physical inquiry evidence



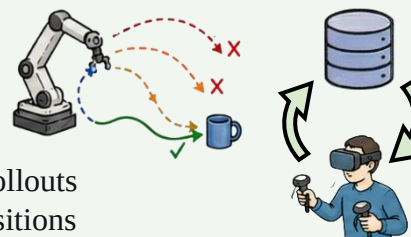
(b) Planning / Decision

- semantic frontiers
- routes / subgoals
- plan repair
- candidate action ranking
- where-to-look potential / value maps



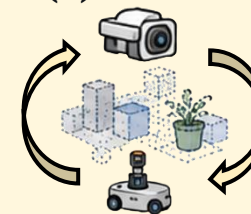
(c) Action

- motor commands
- action chunks
- contact sequences
- human-corrected rollouts
- reset-free task transitions



How: Signals and Mechanisms

(a) Uncertainty-driven



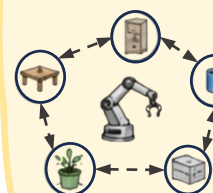
- **Embodied visual exploration:** EVE, ActiveSplat...
- **Physical inquiry / active sensing:** EXCALIBUR...
- **Semantic frontier / object-search ranking:** SemExp, PONI, CoW, VLFM...

(b) Competence-driven



- **Open-ended curriculum:** Voyager...
- **Plan repair / subgoal selection:** DEPS...
- **Test-time action reranking:** V-GPS...
- **Human-corrected RL / policy fine-tuning:** HIL-SERL, DPPO...

(c) Reachability-driven



- **Scene memory / frontier snapshots:** 3D-Mem...
- **Reusable skill libraries:** Voyager...
- **Shared multi-task experience:** MT-Opt...
- **Reset-free task transitions:** Reset-Free Multi-Task RL...